1. 10% of light bulbs produced by a factory are defective. If you buy a box of 5 light bulbs, what is the probability that at least one of the bulbs will be defective?

2. Mean and variance of a binomial random variable are 4 and 2, respectively. Determine $P(X = 0)$.

3. The average number of misprints in one page of a book is 3. What is the probability that there will be exactly 2 misprints in the next page?

4. The average number of people entering a bank between 9:00-10:00 A.M. and 10:00-11:00 A.M. are 3 and 2, respectively. Calculate the probability that at most 2 people will enter the bank between 9:00-11:00 A.M. tomorrow.

5. Automobile battery of a particular brand malfunctions with probability 0.001. Use two different distributions to calculate the probability that at least one out of 1000 batteries will malfunction.

6. A box contains 20 mobile phones that are in good condition and 10 that are defective. You randomly selected 5 phones from the box without replacement. What is the probability that at least 2 of them will be in good condition?

7. Refer to the previous problem. If the box contains 2000 good phones and 1000 defective ones, calculate an approximate probability that at least 2 out of 5 randomly selected phones will be good.

8. A fair dice is thrown until ‘6’ has occurred 3 times. What is the probability that exactly 5 throws will be required?